Kaan Degirmenci

AI CV Link: https://ai-cv-kaan-degirmenci.streamlit.app

Computer Science Student | Future System Architect
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Professional Summary

Forward-thinking Computer Science student evolving from a **traditional coder** into a **system-oriented engineer**. Strong at bridging **low-level computing (C, C++, Assembly, Real-Time Systems, embedded/IoT)** with **high-level software and data architecture (Java OOP, SQL, AI/ML, Python)**. Known for a **first-principles mindset** in AI and an architectural approach to software design, focusing on **scalability, correctness, and end-to-end ownership.**

Education

Frankfurt University of Applied Sciences

- **B.Sc. in Computer Science (Informatik)** | Final Year Student
- *Expected Graduation: Winter 2027*

  GPA: 2.3 (German grading scale)

Selected Coursework & Grades

•Introduction to Programming with **C — 1.0**

•Object-Oriented Programming with **Java – 1.3**

•Object-Oriented Programming with **C++ — 1.7**

•**Analysis – 1,7**

•**Real-Time Systems — 1.7**

•Databases **(SQL) — 2.3**

•**Artificial Intelligence — 2.3**

•**Discrete Mathematics — 2.3**

•**Operating Systems — 2.3**

•Probability & Statistics **(Data Analysis with R) — 2.7**

Technical Skills

Core Engineering Philosophy

- **System Architecture & End-to-End Design**
- Object-Oriented Design **(OOP)**
- **First-Principles AI** & Engineering Logic

Programming Languages

- **Java** (Advanced, OOP, GUI, concurrency)
- **Python** (AI/ML, data processing, RAG)
- **C++ / C** (OOP, Embedded systems, real-time logic)
- **SQL** (Relational databases)
- **R** (Statistical data analysis)
- **Assembly**

Artificial Intelligence & Machine Learning

- **Deep Learning:** CNN architectures, Conv2D & pooling layers, backpropagation
- **Optimization:** Adam, SGD, gradient-based learning
- **Supervised Learning:** Binary vs. multi-class classification (Sigmoid, Softmax), Decision Trees, Random Forests
- **Unsupervised Learning:** K-Means clustering, Elbow Method, Inertia, Silhouette Score
- **Reinforcement Learning:** Agent–environment interaction, MDPs, delayed reward logic
- **Mathematical Foundations:** Linear algebra (matrix operations), calculus (gradients)

Projects

https://github.com/kaanbabaa?tab=repositories

Advanced Traffic Simulation Wrapper (SUMO)

Tech Stack: Java, Object-Oriented Design, GUI (Swing/JavaFX), SUMO Engine

- Engineered a robust **Java wrapper** around the SUMO traffic simulation engine using strict **OOP principles**
- Designed a responsive GUI to **visualize real-time traffic data**
- **Implemented multi-threading** to decouple simulation logic from UI rendering

AI-Powered Technical Interview Engine (Custom RAG)

Tech Stack: Python, Streamlit, ChromaDB, Groq, Pydantic, Sandbox Environment

- **Automated CV & GitHub Profiling:** Engineered an **LLM-driven parser** using Pydantic to convert PDF resumes into structured personas and analyzed GitHub code skeletons to enable project-specific, deep-dive questioning.
- **Intelligent Question Matching:** Built a **custom semantic retrieval logic** to map candidate tech-stacks to a **VectorDB(Chroma)**, ensuring interview questions are precision-matched to individual expertise without using high-level wrappers.
- **Live Sandbox Evaluation:** Integrated a secure execution environment to audit candidate solutions in real-time, leveraging **an LLM to evaluate logic, runtime logs, and Big-O efficiency**.
- **Analytical Reporting:** Delivered a comprehensive dashboard providing detailed interview scoring and **real-time API token usage/cost tracking** for full system transparency.

Smart Trash Bin – IoT System

Tech Stack: C++, ESP8266, Firebase, Google Apps Script

- Designed an **end-to-end IoT architecture** connecting physical sensors to a cloud backend
- **Programmed microcontroller logic** to process HC-SR04 distance sensor data
- Implemented a s**oftware-based signal smoothing algorithm to reduce sensor noise**
- Enabled **real-time data updates via Firebase and WebSocket** communication

Machine Learning & Computer Vision Fundamentals

Tech Stack: Python, NumPy, Custom Neural Networks

- **Built CNN architectures** for multi-class classification (CIFAR-10 Vehicles, MNIST Digits, Kaggle)
- Applied a glass-box approach by manually reasoning about **backpropagation, gradients, and matrix operations**
- Used structured train/test splits to **validate generalization and mitigate overfitting**

AI-Powered CV Assistant (RAG Application)

Tech Stack: Python, Streamlit, Gemini API

- Developed a Retrieval-Augmented Generation **(RAG)** style application to interact with structured CV data
- Designed a **multi-agent architecture** (Visionary vs. Auditor) to reduce hallucinations
- Implemented session-state memory and **modular prompt engineering** for adaptive AI behavior

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Internship Objective

Seeking a Summer 2026 internship that connects low-level engineering (embedded systems, IoT, real-time logic) with high-level software architecture (AI, cloud, data-driven systems). Motivated to contribute beyond task execution by participating in architectural decisions, scalable system design, and real-world engineering problem solving.